

Managers' Body Expansiveness, Investor Perceptions, and Firm Forecast Errors and Valuation

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ABSTRACT

We examine the relation between a measure of managers' physical display—body expansiveness—and favorable reporting practices (in firm forecasts and valuation information) and performance (survival and funding success). We videotape 154 entrepreneurs pitching their business ideas, and use computer vision software to obtain information about speakers' movements. We show that physical expansiveness correlates with higher forecast errors and proposed firm valuations and lower survival rates yet higher likelihood of funding success. We argue that investors may incorrectly interpret nonverbal communication in their assessments of entrepreneurs and propose a behavioral

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explanation. We further corroborate the proposed mechanism by studying investor perceptions of entrepreneurs' personal characteristics. Overall, we shed light on an overlooked source of information—nonverbal behavior—and relate it to firm forecasting, valuation, survival, and financing success, which are important factors in the assessment of investment opportunities, deal structure, and monitoring.

JEL codes: G12, G31, G41, M12, M13

Keywords: investor perceptions; financial projections; valuation; nonverbal communication; entrepreneurs; startups

1. Introduction

Managers communicate information to capital market participants via both quantitative and qualitative information. Accounting and finance research has traditionally focused on quantitative information included in mandatory and voluntary disclosures and, more recently, exploited the textual information in corporate communications. Overall, these studies find *qualitative verbal* communication to be incrementally useful in predicting a variety of firm outcomes. (See Loughran and McDonald [2016] for a survey of the literature.) In contrast, *qualitative nonverbal* information has received less attention (Mayew and Venkatachalam [2012]), despite an abundant literature supporting the important role of nonverbal cues in the dissemination of information (Burgoon, Guerrero, and Floyd [2016]).

We study whether body movement, one type of qualitative nonverbal information, is associated with managers' favorable reporting in firm forecasts and valuation information and with firm performance (survival and funding success). We use computer vision software (OpenPose) to extract two-dimensional skeletal information about managers' physical displays and construct a composite measure of physical expansiveness. Relating individual nonverbal cues to corporate reporting and firm outcomes may seem puzzling. Our arguments rely on the social psychology literature connecting nonverbal behavior to personal characteristics (e.g., Gifford, Ng, and Wilkinson [1985], Borkenau and Liebler [1992], Carney, Hall, and LeBeau [2005], Zebrowitz, Montepare, and Strom [2013]). Nonverbal behavior often arises unconsciously, which explains why individuals can express and maintain a given personality trait without consciously invoking it (Hall, Coats, and LeBeau [2005]). Our study builds on the premise that qualitative nonverbal displays ("expressive behavior") can transmit valuable information about the individual (manager) and thus about firm actions and outcomes (Ambady and Rosenthal [1992], Ambady, Bernieri, and Richeson [2000]).

We use a sample of 154 videotaped entrepreneurs pitching their businesses to an audience of about 100 investors. We obtain a measure of physical expansiveness by adapting a pose estimation program provided by the CMU Perceptual Computing Lab. The program allows us to

extract two-dimensional skeletal information about each speaker (i.e., physical joint coordinates). We further obtain financial forecasts and valuation information from a four-page document that each firm distributes to investors at the beginning of an investment forum. This context suits our research for at least three reasons. First, it ensures that the individual we observe—the entrepreneurs presenting at the pitch contest—has prepared or has been intimately involved in preparing the reported information. In this way, the associations we test between individuals' nonverbal behavior and reported information are much cleaner than in traditional capital market settings, where information likely comes from multiple sources (e.g., departmental forecasts) or is outsourced (e.g., underwriters providing firm valuation numbers). Second, we also test associations between individuals' nonverbal displays and firm survival and funding success. We claim that any individual effect on corporate outcomes is more likely present in early-stage companies, given the crucial role of entrepreneurs in managing all areas of the firm. Third, the investment forums at which the entrepreneurs pitch are homogeneous events from which individuals' body movement can be captured. A comparable event in public companies is not obvious—for example, earnings conference calls are homogeneous, but few firms offer video recordings of them.

Our first research question examines whether entrepreneurs' physical expansiveness is associated with higher deviations between firm forecasts and actual results and with overly high valuations (as proposed by the entrepreneur) compared to similar firms. We base these tests on the social psychology literature, which links physical expansiveness with more dominant, passionate, and attractive individuals (Carney, Hall, and LeBeau [2005], Chen, Yao, and Kotha [2009], Vacharkulksemsuk et al. [2016]), three personal characteristics associated with self-confidence (Hatfield and Sprecher [1986], Mobius and Rosenblat [2006]), assertiveness (Anderson and Kilduff [2009]), feelings of power (Hall, Coats, and LeBeau [2005]), and overconfidence (Fast et al. [2012]). We argue that entrepreneurs with these characteristics make greater forecasting errors and propose higher valuations, which are important informational inputs in the early-stage investment selection process, contract design, and monitoring stage (Kaplan and Strömberg [2003, 2004], Gompers et al. [2020]).

Our second research question explores whether entrepreneurs' physical expansiveness is associated with firm survival and actual investor funding decisions. Note that survival depends on the entrepreneur, whereas funding decisions depend on investors. This distinction matters because it allows the entrepreneur's expansiveness to be directly associated with survival—that is, expansive gesturing conveys individual characteristics that affect firm development—but not with funding decisions, which relate more to investors' perceptions of the entrepreneur. Thus, the relation between the entrepreneur's physical displays and the decision to invest assumes that investor accurately perceive *true* personal characteristics of the entrepreneur (dominance, passion, and attractiveness). We rely on findings

from social psychology to motivate this assumption (Ambady and Rosenthal [1992], Ambady, Bernieri, and Richeson [2000]), and perform additional empirical work to support it further. Overall, we argue that the qualities associated with more dominant, passionate, and attractive individuals—for example, a higher propensity to acquire resources (Keltner, Gruenfeld, and Anderson [2003]) and greater tolerance for risk (Anderson and Galinsky [2006])—can affect firm survival and funding success.

We find that body expansiveness is positively associated with revenues and earnings forecast errors and with overly high proposed firm valuations. These findings are robust to the inclusion of other nonverbal communication measures (which could correlate with body expansiveness), such as voice tone and facial characteristics. Results regarding the second research question show a positive (negative) association between body expansiveness and funding success (survival). We argue that higher confidence and risk tolerance drive expansive entrepreneurs to take more risk, explaining lower survival rates. Even so, investors are more likely to fund this type of entrepreneur. We propose a behavioral explanation consistent with individuals (e.g., investors) attaching a set of positive qualities to passionate, dominant, and attractive people (who display more expansively), even when these qualities do not necessarily translate into better performance (e.g., survival).¹

To further investigate our proposed explanation, we obtain perceptions of entrepreneurs' personal characteristics from a sample of 1,772 independent raters, who watch each pitch and assess speakers on dominance, passion, and attractiveness.² We additionally ask for perceptions of trustworthiness and competence to use as controls. This analysis serves two purposes. First, it supports our main argument that physical expansiveness indeed conveys dominance, passion, and attractiveness. In doing so, we implicitly assume that perceptions of others (the ratings we obtain) accurately capture *true* individual characteristics (dominance, passionate, and attractiveness), an assumption supported by Ambady and Rosenthal [1992] and Ambady, Bernieri, and Richeson [2000]. Second, perceptions per se also matter for our study, especially regarding the decision to invest, which is the only dependent variable relying on perceptions. Ultimately, all what investors really obtain from pitches are their perceptions of entrepreneurs.

¹ An alternative argument would claim that investors invest in highly expansive entrepreneurs in seek of high returns (even at the expense of lower survival rates and higher entry valuations). To the extent that these manager profiles are less risk-averse, they may offer higher growth prospects and therefore better exit opportunities. We provide discussion and empirical work on this alternative argument in the results section.

² Interpreting independent participants' perceptions as professional investor perceptions might seem far-fetched. We argue that the survey questions did not require any financial expertise but merely asked for perceptions of individual personal characteristics (e.g., how attractive someone is). These perceptions of others should be orthogonal to any individuals' professional background or financial literacy.

As is common in behavioral constructs, we obtain five highly correlated variables, and employ principal component analysis to avoid multicollinearity issues. Three components emerge: presence (capturing perceptions of entrepreneur's passion and dominance), attractiveness (loads independently), and professionalism (trustworthiness and competence). Consistent with our main argument, we find that presence is positively and significantly associated with revenue and earnings forecast errors and overly high proposed valuations. Projects run by higher presence entrepreneurs are less likely to survive and yet more likely to be invested. Results on forecasted errors are similar for attractive entrepreneurs. Overall, results from the perception analysis align with the physical expansiveness tests. We argue that body expansiveness transmits information because it is a nonverbal manifestation of greater dominance, passion, and attractiveness, a set of characteristics that affect firm-reporting practices and performance. We believe that this complementary test of perceptions not only corroborates our main argumentation but also reinforces the validity of our physical expansiveness measure.

We further explore cross-sectional variation by the firm's age and venture capital (VC) presence. Both variables likely identify better governance (e.g., management control systems or periodic reporting), which helps the firm cope with organizational complexities or agency frictions associated with the separation of ownership and control (Davila and Foster [2005]). For older and VC-backed firms, we conclude that the effect of individual personal characteristics—captured through physical expansiveness—on favorable firm-reporting practices is mitigated by the discipline provided by stricter governance.

This study contributes to several research streams. First, our study speaks to an accounting audience by showing how managers' body movements can relate to the characteristics of reported financial information and firm outcomes. Second, we add to a growing literature that uses new technology to capture unexplored constructs. To the best of our knowledge, we are among the first to systematically examine body language, bringing new insights to a literature that has mostly focused on facial and voice characteristics (Devine, Holmes, and Wang [2021]). Also, our study incorporates more nonverbal components in one setting than previous studies do: we control for manager's tone of voice, facial characteristics, and verbal content. This suggests that body language likely influences firm outcomes on its own.

Third, we contribute to the literature analyzing nonverbal cues in entrepreneurial settings. Clarke, Cornelissen, and Healey [2019] subjectively code verbal and nonverbal (gesturing) content for 17 entrepreneurial pitches and identify one pitching strategy (the demonstrative approach, based on high gesturing and low figurative language) to be associated with

propensity to invest.³ Hu and Ma [2021] explore facial expression, verbal content, and voice pitch to develop a “pitch factor.” They find that positive pitches (i.e., passionate and warm) increase funding probability despite being negatively correlated with survival, employment, and subsequent VC funding. They explain these findings with investors’ inaccurate interpretations of the positive features of the pitch. We complement their work by focusing on body language—which has received much less attention (Devine, Holmes, and Wang [2021])—and its relation to firm’s financial information.

Finally, our results may be of interest to practitioners, since videos are becoming an accepted form of corporate communication (Cade, Koonce, and Mendoza [2020]). For instance, earnings announcements events (e.g., Netflix, Caterpillar, and T-Mobile) or shareholder meetings are increasingly streamed via online video (Brochet, Chychyla, and Ferri [2021]). Beyond our specific results, an important reflection of our study is that (stakeholder) perceptions of individuals (e.g., CEOs) vary with the content visualized (i.e., character inferences from static facial images will likely differ from those of dynamic body movements). Whether and how these different perceptions later affect individuals’ (e.g., investors’) interpretations and actions remains an open empirical question.

2. *Conceptual Background*

2.1 NONVERBAL BEHAVIOR AND PERSONAL CHARACTERISTICS

Nonverbal communication has long been studied in association with individual characteristics (Zebrowitz, Montepare, and Strom [2013]). This association has captured the attention of social psychologists because nonverbal communication often emerges unconsciously, enabling it to express a personality trait without that trait being explicitly invoked (Ambady and Rosenthal [1992], Hall, Coats, and LeBeau [2005]). For example, Gifford, Ng, and Wilkinson [1985] find that independent raters observing silenced videos of job interviews could accurately rate individual social skills. Three nonverbal cues capture an individual’s social abilities: rate of gesturing, time spent talking, and formality of dress. Borkenau and Liebler [1992] document that strangers infer personal characteristics like extraversion from individual’s voice (e.g., powerful voices), static attributes (e.g., stylish hair or friendly expression), and dynamic behavior (e.g., arm swinging), but they fail to find significant associations for other personal characteristics like agreeableness, culture, and emotional stability. Burgoon, Birk, and Pfau [1990] find that perceptions of competence are inferred from nonverbal cues like greater vocal and facial pleasantness and expressiveness.

³ In contrast, we use an objective measure of gesturing that can be systematically exploited in large data sets. In addition, we examine accuracy of financial forecasts, firm survival, and actual investment, rather than propensity to invest.

Physical expansiveness relates to personal characteristics relevant to our study. Carney, Hall, and LeBeau [2005] document 35 nonverbal cues that are perceived differently between high versus low dominance individuals, with expansive (contractive) gesturing being more common in more (less) dominant individuals. Similarly, Vacharkulksemsuk et al. [2016] argue that more expansive nonverbal displays (e.g., widespread limbs, stretched torsos, or the maximization of occupied space) relate to more dominant and attractive individuals, which translates into a higher likelihood of being chosen in dating events. Chen, Yao, and Kotha [2009] use nonverbal behavior, like energetic body movements, rich body language, or animated facial expressions, to validate a measure of entrepreneurs' passion. More broadly, research on animal behavior finds a robust relation between body expansiveness and dominance behaviors (Ellyson and Dovidio [1985], Hall, Coats, and LeBeau [2005]). From fish to primates, larger bodies reliably predict dominant behaviors in mates (de Waal [1982], Weisfeld and Beresford [1982]).

We use the link between individuals' expansive nonverbal displays and dominance, passion, and attractiveness to theoretically ground our research questions. Note, however, that the only outcomes we can directly link to entrepreneurs' expansive displays—and the associated personal characteristics—are firm forecast errors, valuation, and survival, as they ultimately depend on the entrepreneur we observe. In contrast, the decision to invest depends on the perceptions investors form about entrepreneurs. This raises the question of whether physical displays can also inform (investor) perceptions of others (entrepreneurs), in which case the effect of expansiveness on the decision to invest can be justified. We again rely on social psychology, which establishes that perceptions of others accurately capture personal characteristics—dominance, passion, and attractiveness—in observations of individual's dynamic displays (Ambady and Rosenthal [1992], Ambady, Bernieri, and Richeson [2000]).⁴ Kogan et al. [2011] emphasize this accuracy by showing that individuals display or behave in a certain manner (e.g., dominant individuals displaying more expansively) following innate biological factors. The authors show that independent raters successfully perceive more prosocial behavior in individuals carrying the G allele of the oxytocin receptor gene (shown to induce more prosocial abilities) than in those carrying the A allele.

Studies closer to our context corroborate the argument that perceptions can relate to actual outcomes. Positive assessments of entrepreneurs

⁴ Studies from Ambady and coauthors rely on individual's dynamic physical displays (i.e., body motion) to conclude that perceptions of individuals' characteristics accurately approximate such characteristics. This link, however, may not be robust to perceptions extracted from other nonverbal cues, like static facial images (e.g., Todorov et al. [2005] or Todorov et al. [2008]). Seemingly, the body itself and the dynamic (rather than static) context can lead to more accurate inferences of others (Ambadar, Schooler, and Cohn [2005], Aviezer, Trope, and Todorov [2012]).

have been associated with stronger firm performance four years later (Huang and Pearce [2015]), higher perceptions of borrower trustworthiness with better firm credit scores and lower default rates (Duarte, Siegel, and Young [2012]), and perceived borrower attractiveness with a higher likelihood of receiving a loan and subsequent defaults (Ravina [2019]).

2.2 PERSONAL CHARACTERISTICS AND FIRM ECONOMIC OUTCOMES

Overall, the social psychology literature argues that nonverbal cues can transmit valuable information about individual characteristics. With respect to managers, these characteristics can have important implications for the firm. Bertrand and Schoar [2003] highlight the importance of manager “fixed effects” (i.e., specific individual characteristics) in corporate investing, financing, and strategic decisions. Subsequent work has examined the effect of managers on investment decisions (Malmendier and Tate [2008]), tax avoidance (Dyreng, Hanlon, and Maydew [2010]), voluntary disclosure (Bamber, Jiang, and Wang [2010]), stock return variability (Adams, Almeida, and Ferreira [2005]), financial reporting risk (Davidson, Dey, and Smith [2015]), and tone in regulatory filings and conference calls (Davis et al. [2015], Hendricks, Lang, and Merkley [2021]). Our study explores entrepreneurs’ characteristics associated with physical expansiveness—dominance, passion, and attractiveness—and relates these to firm-reporting and performance outcomes.

The evidence on the association between dominance and firm outcomes is mixed. A dominant individual aims at gaining or maintaining influence over others (Mast and Cousin [2013]) and is perceived as more assertive, proactive, and confident than peers (Anderson and Kilduff [2009], Vacharkulksemsuk et al. [2016]). The importance of dominance in a financing context stems from the fact that more dominant individuals are more likely to acquire resources (Keltner, Gruenfeld, and Anderson [2003]) and initiate negotiations in competitive scenarios (Magee, Galinsky, and Gruenfeld [2007]). The literature also links dominance to higher feelings of control (Fast et al. [2009]) and willingness to act (Galinsky, Gruenfeld, and Magee [2003]). Although these findings suggest dominant individuals could boost firm economic outcomes, evidence also supports opposing arguments. Fast et al. [2012] find that overconfident decisions are more common among dominant people and that these decisions are more likely to lead to monetary losses. Similarly, Malmendier and Tate [2008] find that overconfident CEOs overestimate their ability to generate returns, overpay for acquisitions, and undertake value-destroying mergers. Anderson and Galinsky [2006] observe that dominance (sense of power) increases optimism and risk taking. Overall, dominant individuals appear to contribute both positively and negatively to firm outcomes.

Similar to dominance, passion can also lead to either positive or negative outcomes for the venture. Murnieks et al. [2016] document that it relates to tenacity, inspirational leadership, and dedication to the venture. But passion can also lead to obsessiveness or too narrow a view of the venture

or activity in question (Vallerand et al. [2003]). This narrow scope has also been found in forecasting: Kahnemann and Lovallo [1993] conclude that these types of individuals rely too much on the specifics of a case at hand (i.e., details of the venture and the obstacles to overcome) rather than on base rate information (e.g., objective predictors). This phenomenon has been termed the “inside view” and can limit the search for information, cloud decision making, and lead to *rosy* forecasts (Kahneman and Lovallo [1993], Cassar [2010]).

Attractiveness, too, figures in a variety of corporate contexts. Brooks et al. [2014] document that attractive males are more likely to receive investments than nonattractive ones. Blankespoor, Hendricks, and Miller [2017] find a positive and significant association between attractiveness and IPO valuations. Relatedly, a recent study in a peer-to-peer loan granting context reports that more attractive individuals are more likely to obtain a loan (Ravina [2019]). Also, Baugh, Hallman, and Kachelmeier [2022] document that less-expert audit committees hire more attractive audit partners and pay more for their services. Psychological studies posit that some of the positive outcomes related to attractive people stem from childhood, with educators having higher expectations for attractive children and therefore devoting more time to them, fostering their confidence and social abilities (Hatfield and Sprecher [1986], Mobius and Rosenblat [2006]). On the other hand, several studies report the presence of a beauty premium (or a physical attractiveness stereotype), by which people overestimate the abilities of more attractive individuals (Eagly et al. [1991], Feingold [1992]), even when their performance has been shown to be comparable or worse to that of less attractive peers (Mobius and Rosenblat [2006], Graham, Harvey, and Puri [2017], Ravina [2019]). Mobius and Rosenblat [2006] use a labor market experiment and find that employers *wrongly* expect good-looking candidates to outperform less attractive ones and hence provide them higher compensation. The study finds that attractive people do not outperform the less attractive. Ravina [2019] even documents that attractive individuals are more likely to default on their loans. Overall, this evidence is inconclusive regarding how attractiveness might affect economic outcomes.

We argue that physical expansiveness conveys information about personal characteristics (dominance, attractiveness, and passion) that can affect important firm outcomes. To better identify the effect of physical expansiveness, we also address trustworthiness and competence, as these characteristics can be relevant in financing contexts. For example, trustworthiness has been associated with success in crowdfunding campaigns (Duan et al. [2020]), higher probabilities of receiving a loan and lower interest payments (Duarte, Siegel, and Young [2012]), and lower audit fees charges (Hsieh et al. [2020]). These findings are consistent with trustworthy appearances signaling higher reputational capital, which translates into positive firm outcomes (Duarte, Siegel, and Young [2012]). Regarding competence, Graham, Harvey, and Puri [2017] document that, despite not showing better performance, more competent-looking CEOs receive

higher compensation and are placed in bigger firms. Another study documents that more competent-looking individuals are more likely to win elections (Todorov et al. [2005]). Together, these findings support the idea that a competent appearance can shape perceptions of others.

2.3 NONVERBAL BEHAVIOR AND MACHINE LEARNING STUDIES IN ACCOUNTING AND FINANCE

Over the past two decades, scholars have exploited advances in textual analysis to investigate market implications of *qualitative verbal* information in such corporate communications as annual reports, financial news, communication with regulators, or earnings announcements. Yet, there exists limited evidence on the market effects of *qualitative nonverbal* information (Zebrowitz, Montepare, and Strom [2013]). This is mostly because of unstructured data sources and the lack of appropriate software.

Body language has been especially overlooked. Devine, Holmes, and Wang [2021] provide a recent review of individual aesthetic attributes by screening 65 papers: 54 are devoted to facial features, seven to voice, and just two to body language. (The other two explore managers' height and physical fitness.) These statistics are somewhat reflected in the accounting and finance literatures, which study facial characteristics considerably more. Jia, van Lent, and Zeng [2014] use ImageJ software to extract information about CEOs' facial width-to-height ratio (*fWHR*), a measure that reflects CEO masculinity, including aggression, egocentrism, and hunger for social status. The authors document a positive association between *fWHR* and financial misreporting. He et al. [2019] use the same software to report a positive association between analysts' facial characteristics and forecast accuracy. Hsieh et al. [2020] use machine learning algorithms to construct a manager facial-trustworthiness measure and document that auditors charge lower audit fees to firms with trustworthy-looking CFOs. Similarly, Duan et al. [2020] use the same measure to report a positive association with crowdfunding campaign success. Hobson, Mayew, and Venkatachalam [2012] and Mayew and Venkatachalam [2012] analyze voice tone (through Layered Voice Analysis software) and examine conference call audio files. Overall, they find that managerial vocal cues contain incremental (to quantitative and textual) information about firm's financial future and the likelihood of misreporting. In general, studies focus on just one nonverbal component.

We address multiple attributes of nonverbal components in this study: we control for manager's tone of voice, facial characteristics, and verbal content. Doing this enables us to conclude that body language can significantly influence firm outcomes on its own. A similar study by Choudhury, Wang, Carlson, and Khanna [2019] uses Microsoft Azure facial recognition software (nonverbal) and topic and sentiment analysis (verbal) of CEO interviews to identify five managerial communication styles. But theirs is a mainly methodological study that provides a proof-of-concept analysis of

the different dimensions of CEO communication in mergers and acquisitions (M&A) settings.

Closer to our paper is the work of Hu and Ma [2021], who analyze three nonverbal dimensions—facial, vocal, and verbal—in a sample of entrepreneurs' pitches posted to YouTube or Vimeo. They depict the visual component using facial expression analysis (Face++), the vocal component using voice analysis (Python packages), and the verbal component employing standard linguistic dictionaries. They conclude that entrepreneurs scoring high in positive dimensions (e.g., individual arousal or valence) are more likely to obtain funding but perform poorly (lower headcounts, survival, funding amounts, and future funding success). Although the use of commercially available software offers the advantage of lower computational burdens and higher ease of reproducibility, the authors are constrained to the output featured by the software.

Our paper contributes to this literature and uses open source code to extract individuals' body movements from a full sample of presenters at an important European early-stage investment network. To extract individuals' body movements, we use OpenPose, a pose estimation algorithm developed by the CMU Perceptual Computing Lab that extracts two-dimensional skeletal information from video inputs. The analysis requires some computational expertise but suits our research needs while facilitating reproducibility, thanks to the open source code. Different from other scholars, we also complement our main analysis with subjective measures of entrepreneurs' personal characteristics.

2.4 FINANCIAL FORECASTS, FIRM VALUATION, AND EARLY-STAGE EQUITY FINANCING

Early-stage investors face severe (*ex ante*) information asymmetries and (*ex post*) agency conflicts in their investment decisions. Financial contracting theory provides three means to mitigate these conflicts (Kaplan and Strömberg [2001]). First, investors can collect information to screen out bad projects/entrepreneurs (screening stage). Second, they can write contracts that provide incentives for entrepreneurs to behave optimally (contracting stage). Third, upon investment, they can use information collection and monitoring to enforce the contract (monitoring stage).

Although financial forecasts and valuation information matter in all three stages, we focus on screening. Gompers et al. [2020] point that the two most important metrics for VC investments are the multiple over invested capital (MOIC) and the internal rate of return (IRR). Both require valuations and forecasts, which are provided in the business plans we observe. Furthermore, the same authors report that 69% of early-stage investors use financial projections in their investment assessments and 63% use available capital and target ownership as a shortcut to evaluate entrepreneurs' proposed valuations. Another related study is by Kaplan and Strömberg [2004], who document that after the quality of the management team, high valuations and unrealistic financial projections are

the second (out of six) internal factors investors consider when assessing investment weaknesses.

Firm forecasts and proposed valuations are also highly relevant in the contracting and monitoring stage. Kaplan and Strömberg [2003] document how specific contracting features, such as board, voting, and cash flow rights, can depend on revenue and earnings targets or valuation information. For instance, if the firm does not meet revenue targets, the VC firm can obtain additional board seats and voting power (board and voting rights) or limit the entrepreneur's stock option compensation (cash flow rights). Valuation information is also used for paying stock dividends to VCs if new funding is available above a certain valuation (voting and cash flow rights).

The above findings comport with financial contracting theory, which predicts a higher use of performance-based measures—such as financial targets and valuation numbers—in settings with high information asymmetry and agency conflicts (Kaplan and Strömberg [2004]). Our paper seeks to uncover an overlooked information set—managers' nonverbal displays—by relating it to important metrics in early-stage investments (i.e., firm forecasts and valuation information).

3. Data

3.1 SAMPLE AND SETTING

Our population begins with 245 entrepreneurial teams pitching their startups to an audience of about 100 investors between the years 2012 and 2015. These pitches occur at bimonthly investment forums organized by a European Business Angel and Venture Capital Network. Each meeting usually features between six and eight 10-minute uninterrupted presentations.⁵ This community is a network of active business angels (BAs) and VCs in Europe and convenes entrepreneurs seeking funding and early-stage investors. To present at the forum, startup managers fill out a standardized company profile that investors receive immediately before the forum. The profile includes general information about the startup (e.g., description of the business and founding dates) and a series of open-ended questions: target market, technology, distribution and marketing, key clients, strategic alliances, main competition, major milestones, strategic goals, and management team details. In addition, the profile provides information on previous funding rounds (if any), main shareholders, business forecasts, and a summary of the current investment round (e.g., targeted amount to be raised, expected valuation, and usage of

⁵ An informal committee ensures that projects are not too early (just a PowerPoint presentation) or late stage (already an established company) to meet investors' interests. Also, several industries are automatically discarded (e.g., arms, sex). As usual in these contexts, there can be selection toward internet-based projects.

TABLE 1
Final Sample

Panel A. Sample selection	Firms	Firm-Year Observations
Videotaped projects (from April 2012 to December 2015)	245	
Bad recordings	−5	
Accounting information not available (BvD)	−57	
Revenue and earnings not available (actual or forecasts)	−29	
Number of firms	154	441
Panel B. Industry distribution (primary NACE codes)	Firms	%
Administrative and support activities	10	6.45
Information and communication	52	33.77
Manufacturing	13	8.44
Professional, scientific and technical activities	54	35.06
Wholesale and retail	25	16.23
Total	154	100.00

Panel A describes our sample composition. Our study presents analyses performed on a sample of 154 firms (and 441 observations in specifications with several years of forecasted data per firm). Panel B details our sample composition by industry based on primary NACE codes. Appendix B describes all variables.

funds). Appendix A and online appendix 1 provide examples of company profiles.

The forum begins with investors receiving the company profiles. After a brief introduction from the organizers, the presentations take place. Sometimes brief Q&As follow, but they are not systematic across forums (e.g., time constraints) and, when available, recordings are inconsistent. This prevents us from analyzing this more spontaneous part of the pitch. After the session, all participants can attend an informal social event. We do not have any data on this event.

Panel A of table 1 describes the sample composition. We discard five projects out of the original 245 because of serious recording deficiencies (e.g., entrepreneur is not displayed). We then merge projects with accounting information provided by Bureau van Dijk’s database. We lose 57 projects for which the database has no official financial statements. In addition, we could not obtain either actual or forecasted valid information on revenues or earnings for 29 startups, leaving our final sample at 154 firms (441 firm-year observations). In terms of industry composition (see panel B of table 1), most startups are Internet-based and operate mainly in the information and communication (e.g., aggregators or marketplaces) or the scientific and technical sectors (e.g., medical or computer programming companies).

3.2 PHYSICAL EXPANSIVENESS AND NONVERBAL MEASURES

To obtain the physical expansive measure, we use a pose estimation program provided by the CMU Perceptual Computing Lab.⁶ We obtain 154 videos featuring entrepreneurs pitching their projects. Panel A of figure 1 provides three examples of the setting.

Each video includes one project presentation but may include more than one speaker (out of 154 projects, 17 are presented by more than one speaker). In those cases, we manually segment videos to obtain the physical expansiveness measure per speaker. Next, we extract two-dimensional skeletal information about each speaker (i.e., the X-Y physical joint coordinates). To ease the extraction process, the CMU Perceptual Computing Lab provides a github repo, including instructions on how to compile the source code into binary executables that we use. Panel B of figure 1 illustrates the raw code we use to open a video file, pass it through OpenPose, and write an output file per frame containing each physical joint coordinate. The code creates a new video file with a skeleton layer superimposed on the individual.

Panel C of figure 1 provides an excerpt of an output video file. As observed (middle image), the software detects any human pose, including those of members of the audience. We manually check each video and, when necessary, require OpenPose to read individuals who are facing the camera, centered in the frame, and having more joints captured (as compared to other skeletons in the frame). In addition, we also need to edit some missing joints—very few frames—by manually filling coordinate positions; we impute midpoint values between the joint coordinates of the previous and posterior frame. The right image of panel C depicts the outcome of our manual editing. We obtain around 18,000 JSON files per project (i.e., one file per frame, at 30 frames per second, on an average 10-minute video). These files provide two-dimensional coordinates (X and Y) per frame that position each body joint in a two-dimensional pixel space.

To capture body expansiveness, we focus on the movement of body extremities (hands and feet) and the head. In particular, we compute the distance between two consecutive positions (i.e., dynamic movement) for each physical joint (extremity). Because these distances are minimal within two consecutive frames—there are 30 frames per second—we opt to compute distances every other 15 frames (i.e., every half a second). Thus, we run a Python script to extract coordinates information per joint every 15 JSON files and then compute distances (in pixels) between two consecutive (half a second) frames in the two-dimensional space. We normalize those distances using the speaker's height (i.e., distance from neck to hip). Panel D of figure 1 illustrates this process by plotting the distance (Y axis) between

⁶ OpenPose source code is publicly available at the following github repo: <https://github.com/CMU-Perceptual-Computing-Lab/openpose> (last accessed: October 28, 2020). It follows from work in Cao et al. [2021].

Panel A. Illustration of three entrepreneurs at the investment forum event



Panel A provides three examples (entrepreneurs) to illustrate the investment forum event. We obtain 154 videos displaying entrepreneurs for about 10 minutes. We start the physical display extraction process from these input videos.

Panel B. Raw code to initiate the physical displays extraction process

```
openpose.bin --video <input_path>/video.mp4 --write-json <output_path>/json --write-video <output_path>/openpose.avi --display 0; done
```

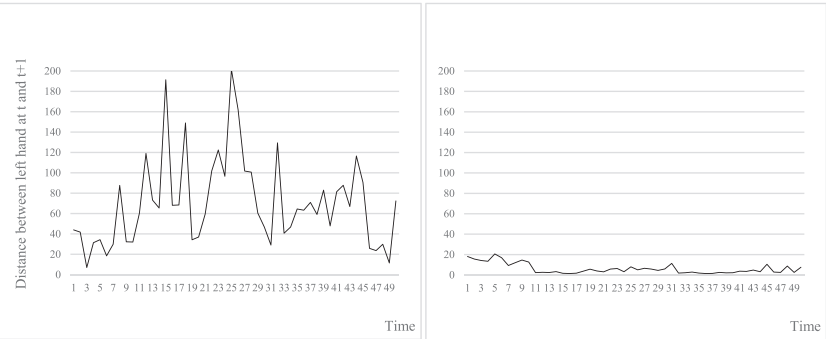
Panel B provides the raw code that opens a video file (mp4 format), passes it through OpenPose, and write an output file per frame with each physical joint coordinate (i.e., one .json file per frame). It also creates a new video file with the skeleton overlaid to the individual (.avi). We ask not to display such .avi video on screen (--display 0 option).

Panel C. Reading human joints on input videos using OpenPose



Panel C provides the video output from the pose estimation procedure. Because our aim is to capture body expansiveness, we require coordinates X and Y (physical joint positions) of each speaker's hands, feet, and head (i.e., nose physical joint). We extract information at the first frame of every half a second (there are 30 frames per second). More specifically, we retrieve information on the movement of left/right hands and feet, and the head.

Panel D. Illustration of left-hand movement over time



Panel D illustrates distances (in number of pixels) between two coordinates at time t and $t+1$ (i.e., half a second later) for the left-hand physical joint. Left (right) graph represents a high (low) expansive speaker. We normalize all distances using the speaker's height (i.e., distance from neck to hip). Sharp spikes represent more expansive physical displays (i.e., greater shifts of the left hand within half a second time). Our measure of expansiveness averages all distances per project over the time series.

FIG. 1.—Physical expansiveness.

TABLE 2
Physical Expansiveness Principal Component Analysis

Panel A: Correlations for physical display variables					
	1	2	3	4	5
Left foot	1.00				
Right foot	0.98	1.00			
Left hand	0.59	0.61	1.00		
Right hand	0.57	0.57	0.81	1.00	
Head	0.54	0.54	0.82	0.80	1.00

Panel B: Principal component analysis			
	Eigenvalue	% of Variance Explained	Cumulative % of Variance Explained
Component 1	3.71	0.74	0.74
Component 2	0.84	0.17	0.91
Component 3	0.24	0.05	0.96
Component 4	0.18	0.04	1.00
Component 5	0.21	0.00	1.00

Table 2 describes the principal component analysis performed for the physical expansiveness construct. We obtain a coordinate location (X, Y) of each physical joint at every half a second, and compute the distance (in number of pixels) between two consecutive coordinates t and $t+1$. We then average those distances to obtain a measure per project and physical joint. A detailed explanation of the extraction process is provided in section 3.2 of the paper. Panel A shows correlations among speakers’ physical joints (all significant at a 1% level). Panel B presents the principal component analysis main output. We use the first component (eigenvalue higher than 1) as our measure of body expansiveness. When using the 137 presentations with only one presenter, statistics are similar to those in panels A and B.

two consecutive left-hand positions over time for a high (left) and low (right graph) expansive speaker. Intuitively, sharp spikes represent greater shifts of the left hand and therefore more expansive gesturing.

Finally, we create a composite measure of body expansiveness by averaging all those normalized distances per project and physical joint. As expected, very high correlations among physical joints are observed (see panel A of table 2); that is, expansive speakers use ample gesticulation in all their body extremities simultaneously. We use a principal component analysis to compile the five variables into one component—the only one reporting an eigenvalue greater than 1—that captures overall body expansiveness (see panel B of table 2). Notably, we have 17 projects that are pitched by more than one speaker, in which case, we average every speaker’s distances per physical joint.

Besides body expansiveness, we measure additional components of non-verbal communication. First, we capture speakers’ voice pitch. We extract the audio component from our original pitch video files (Python ffmpeg package) and then pass it through Praat, which extracts voice fundamental frequencies.⁷ We configure the parameters following Dietrich, Hayes and

⁷ Praat software implements a variation of the Boersma [1993] algorithm. It is available at <https://www.fon.hum.uva.nl/praat/>

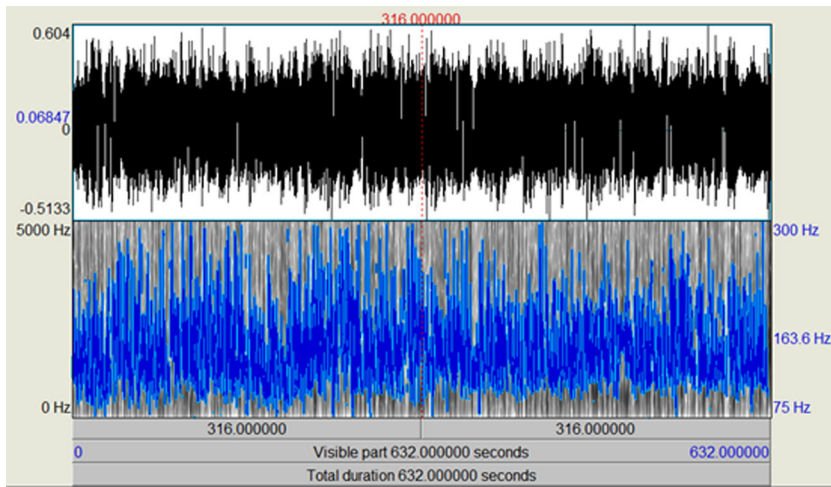


FIG. 2.—Voice pitch analysis: sound waveform, spectrogram, and voice pitch.

The figure provides an example of an audio sound wave (upper chart) and its respective spectrogram and pitch analysis (bottom chart). We are interested in the pitch analysis (blue layer in the bottom chart), which outputs speaker's voice fundamental frequencies (i.e., range of sound frequencies for a given audio signal, measured in Hertz) over the entire audio file. For instance, at second 316 (out of a 632 second audio file), the speaker's voice fundamental frequency displays 163.6 Hz. We extract the mean fundamental frequency for each audio file using Praat (an audio software) and follow Dietrich, Hayes and O'Brien [2019] to configure the input parameters: pitch floor at 75 Hz (100 Hz) and pitch ceiling at 300 Hz (500 Hz) for male (female) audio signals. The remaining input variables (e.g., window length, window shape, voicing thresholds) are set according to Praat's default settings.

O'Brien [2019]—that is, pitch floor at 75 Hz (100 Hz) and pitch ceiling at 300 Hz (500 Hz) for male (female) voices—and use the Praat default settings for the other three parameters (window length, window shape, and voicing threshold). With this, we obtain a mean fundamental frequency value per speaker, which we use as control in all our specifications. Figure 2 illustrates the voice pitch extraction process.

Second, we also capture the *fWHR* through our own analysis (i.e., physical joints extracted from OpenPose). Numerous studies suggest the importance of individual facial characteristics to a variety of firm outcomes (e.g., Jia, van Lent, and Zeng [2014], He et al. [2019], and Hsieh et al. [2020]). Third, we follow Choudhury et al. [2019] to capture pitch sentiment. We employ the R package Syuzhet and define *Pitch_sent* as the number of negative words over the total negative and positive words (i.e., values closer to 1 indicate more negativity and vice versa).

3.3 SUBJECTIVE MEASURES OF PERSONAL CHARACTERISTICS

To obtain measures of perceived personal characteristics, we asked independent respondents to rate the 154 videotaped entrepreneurs' presentations on five personal characteristics: passion, dominance,

attractiveness, trustworthiness, and competence. The first three variables have been associated with expansive physical displays; the latter two are captured to control for potential confounding effects. Following Blankespoor, Hendricks, and Miller [2017], we used content-filtered videos to isolate investors' perceptions of entrepreneurs from the verbal information conveyed throughout the presentation.⁸ Low-frequency filters mumble the voice so content becomes uninterpretable and the voice itself becomes noise. Thus, we capture a behavioral measure of perceived personal characteristics unrelated to the verbal content of the presentation (e.g., growth prospects, strategic alliances, or past performance), which also affects investor perceptions of the entrepreneur. Next we cut videos into 40-second clips that we then present to 2,860 participants via the Turk Prime platform. Research in social psychology validates the use of thin sliced videos to represent an entire behavioral sequence (Ambady and Rosenthal [1992], Ambady, Bernieri, and Richeson [2000]), and more recent studies rely on this technique to validate perceptions of management (Blankespoor, Hendricks, and Miller [2017]).

The survey includes four sections that respondents completed in about nine minutes on average. We first collect a battery of demographic characteristics and then randomly assign one video (out of the 154) as a test video. We ask respondents to adjust sound levels and familiarize themselves with the survey's flow. Next we provide a sequence of five randomly assigned videos and, to reduce concerns related to systematic within-participant response bias, require only three (out of five) randomly assigned personal characteristics (seven-point Likert scale), which appear displayed in random order. Finally, several attention check questions validate the quality of the responses. This process leaves 1,772 valid unique respondents, which is equivalent to an average of about 30 ratings per speaker and personal characteristic. Online appendix 2 includes the survey instrument.

The output from the survey generates five highly significant correlated behavioral variables. Thus, we reduce the dimensionality using principal component analysis. (Online appendix 3 describes the process in detail.) The analysis lends itself to three components. Items that load on the first component are dominance and passion, two traits that can comprise presence (Cuddy, Kohut, and Neffinger [2013]), and so we name this factor "Presence." Attractiveness loads alone in the second component, so we name this component "Attractive." Finally, the third component combines trustworthiness and competence, attributes associated with diligence and professionalism—we name this component "Professional."

⁸ We use the Audacity software to content-filter videos. We use LowPass frequency filters at 200 Hz and 36dB octaves to filter the voice. When required, we manually modify those thresholds and amplify the volume.

3.4 DEPENDENT VARIABLES

Our dependent variables are revenue and earnings forecast error (*Forecast_Rev_Err* and *Forecast_Inc_Err*, respectively) and high firm proposed valuations (*Val_indic*). We use an approach similar to that of Hribar and Yang [2016] and compute the difference between the forecast and actual revenue or earnings number and scale it by size (total assets). The idea is that a deviation of €5,000 in bigger companies (high asset values) shows a relatively good forecast (i.e., relatively small prediction error), whereas the same deviation in smaller companies reflects a rather bad prediction. Unsubstantiated statistics indicate the presence of several outliers in both forecast error measures, and hence we split them into five quantiles (from 1—more pessimistic forecasts—to 5—more optimistic). We propose alternative measures of forecast errors in online appendix 4.

Next, we examine entrepreneurs' firm proposed valuation (as reflected in the documentation investors receive at each forum). In particular, we seek to identify those valuations that appear to be too high (*Val_indic* = 1). We acknowledge that determining whether a startup is under- or overvalued is challenging, so we compare the proposed valuation against a benchmark. We set the benchmark to be the in-sample mean of the ratio between firm valuation (the valuation amount) and revenues by financing round. Expressing the valuation amount as a multiple of the firm's revenues is meaningful in a startup context because revenues generally validate the product and the company business model more broadly. If a given project shows a valuation over revenues higher than the mean for a given round, we classify that observation as highly valued. (See panel C of table 3 for a financing round summary.) We also report results on industry (instead of round) peer comparison in online appendix 4.

Finally, we study firm survival and funding success. The national accounting register reports the status of a company in its last available annual filings. We construct an indicator variable that equals 1 for active companies (111 out of 154) and 0 for inactive ones (i.e., dissolution, insolvency procedures, winding up, or provisional deregistration in the Bureau van Dijk's data set).⁹ We also obtain the investment (funding success) outcome from the business angel and VC network.

3.5 DESCRIPTIVE STATISTICS

Table 3, panel A, presents descriptive statistics for time-invariant variables and first-year observations for *Forecast_Rev_Err*, *Forecast_Inc_Err*, and *Employees*. Both revenue and income forecast errors are measured between 1 (more pessimistic forecasts) and 5 (optimistic). The mean for the one-year-ahead sample is 2.71 and 2.28, respectively (panel A), and 3.93 and

⁹ We manually inspect those companies deemed inactive by Bureau van Dijk. We correct 10 misclassifications or missing values when we find indications of activity (e.g., active Web site, activity on social platforms, or active LinkedIn profiles) or acquisition events. We use Internet searches and the Bureau van Dijk database, which provides ownership data.

TABLE 3
Descriptive Statistics

Panel A: Descriptive statistics (time invariant and first-year observations)									
Variable Name	N	Mean	Std. dev	Min	Max	p25	p50	p75	
Expansiveness	154	0	1	-1.047	7.653	-0.576	-0.233	0.287	
Expansiveness_Ispeak	137	0	1	-1.032	7.332	-0.577	-0.274	0.254	
Forecast_Rev_Err	154	2.714	1.130	1	5	2	3	3	
Forecast_Inc_Err	154	2.279	1.265	1	5	1	2	3	
Val_indic	136	0.140	0.348	0	1	0	0	0	
Active	154	0.721	0.450	0	1	0	1	1	
Invest	154	0.260	0.440	0	1	0	0	1	
Presence	154	0	1	-3.567	2.129	-0.604	0.028	0.638	
Profession	154	0	1	-2.517	2.182	-0.641	-0.015	0.726	
Attractive	154	0	1	-2.655	3.218	-0.712	-0.012	0.525	
Firm age	154	32.701	31.143	0	243	16	27	37	
Employees	154	4.994	5.009	1	38	2	3	7	
Venture Capital	154	0.143	0.351	0	1	0	0	0	
Business Angels	154	0.481	0.501	0	1	0	0	1	
Accelerator	154	0.084	0.279	0	1	0	0	0	
Debt	154	0.156	0.364	0	1	0	0	0	
Log(Capital)	154	12.602	0.788	10.820	15.607	12.206	12.612	13.122	
Round number	154	1.221	1.056	0	5	0	1	2	
Male ratio	154	0.819	0.243	0	1	1	1	1	
Industry experience ratio	154	0.600	0.411	0	1	0	1	1	
Startup experience ratio	154	0.274	0.319	0	1	0	0	1	
Pr_voice pitch	154	143.71	31.34	76.12	246.59	124.95	138.76	155.11	
Pr_fWHR	154	1.163	1.388	0.051	8.219	0.684	0.805	0.975	
Pitch_sent	154	0.213	0.065	0.029	0.364	0.169	0.215	0.256	
Av_fest_rev_g	154	64.759	494.644	0.543	5,979.489	2.285	3.942	7.515	
Breakeven	154	2.195	1.200	0	5	2	2	3	

(Continued)

TABLE 3—(Continued)

Panel B: Descriptive statistics (observations second year and more)									
Variable Name	N	Mean	Std. dev	Min	Max	p25	p50	p75	
Forecast_Rev_Err	287	3.927	1.044	1	5	3	4	5	
Forecast_Inc_Err	287	3.808	1.244	1	5	3	4	5	
Employees	287	8.990	10.904	1	103	3	5	12	

Panel C: Descriptive statistics (rounds of funding)			
Variable name	Frequency	Percentage	Cumulative
Round 0	53	34.42	34.42
Round 1	28	18.18	52.60
Round 2	63	40.91	93.51
Round 3 or more	10	6.50	100.00
Total	154	100	

Panel A of table 3 reports statistics from time invariant variables taken at forum date (e.g., having been invested by a VC, BA, or accelerator at forum date), individual characteristics (e.g., physical expansiveness or ratings on perceptions of entrepreneurs’ personal characteristics). Panel A also shows first-year observations of time variant variables (e.g., first-year forecast errors), whereas second or more years are summarized in panel B. Val_indic shows only 136 observations because of some projects not reporting their company valuations. Panel C provides statistics on the distribution of funding rounds at forum date. Appendix B describes all variables.

3.81 for the longer term forecasts (panel B). This statistic comports with more positive forecast errors (optimism) observed in longer time horizons where information is less readily available.

The third outcome variable of interest relates to the (high) proposed firm valuation (*Val_indic*); a mean of 0.14 indicates that 14% of entrepreneurs present a valuation (over revenues) above the mean valuation for a given round (*Val_indic* = 1). This variable is a conservative measure of high valuations, because it only considers the few firms (14%) that *truly* propose overly high valuations relative to comparable firms. As for survival, we show that 72% of the firms are still alive. Finally, the investment indicator variable shows that 26% of the companies received financing after the forum. Regarding individual characteristics variables, all measures (*Expansiveness*, *Presence*, *Attractive*, and *Profession*) are standardized and behave similarly. Voice pitch (*Pr_voice_pitch*) averages at 143.7 Hz, and the pitch sentiment mean equals 0.213, which indicates, as expected, that the mean pitch has a positive tone (i.e., closer to 0).

We also include a set of control variables: firm age at forum date (mean and median of 32.7 and 27 months, respectively), size (the average startup has around five employees at forum date), presence of prior outside financing (i.e., business angels, VC, accelerators, or lenders), amount of capital requested, and financing round. We also control for team characteristics, such as proportion of males in the entrepreneurial team and percentage of members with prior industry or startup experience. Finally, we add two additional controls in the *Invest* specification. Because the decision to invest can depend on estimated MOIC and IRR, two statistics we do not observe but that depend on forecasted information, we include average forecasted revenue growth (*Av_fcst_rev_g*) and profitability (*Breakeven*), the latter capturing the number of forecasted years before experiencing profits.

Table 4 summarizes main correlations. *Expansiveness* is positively and significantly correlated with *Presence* and *Attractive*, *Val_indic*, and *Pr_voice_pitch* (presenter's voice fundamental frequency). We are, however, cautious about concluding that biological characteristics generally correlate—with more expansive individuals showing slightly higher frequency voices—given that neither *Pr_voice_pitch* nor *Expansiveness* show significant correlations with facial characteristics (*Pr_fWHR*). The latter suggests that body language may operate independently from other components of nonverbal communication. Other notable correlations are those between forecasts of annual revenue growth at pitch date (*Av_fcst_rev_g*) and revenue forecast errors (*Forecast_Rev_Err*) or overly high valuations (*Val_indic*), which suggest that more demanding projections (which comes with overly high valuations) are not later realized.

TABLE 4
Correlation Table

N= 154		1	2	3	4	5	6	7	8	9	10	11	12	13
1	Expansiveness	1												
2	Forecast_Rev_Err	0.061												
3	Forecast_Inc_Err	0.034	0.070											
4	Val_indic	0.147*	0.138	0.036										
5	Active	-0.071	-0.042	-0.046	0.189**									
6	Invest	0.024	0.032	0.186**	-0.001	0.006								
7	Presence	0.216***	0.047	0.157*	0.014	-0.153*	0.196**							
8	Profession	0.059	0.059	0.038	0.088	0.095	0.027	0.000						
9	Attractive	0.189**	0.164**	0.106	0.002	-0.051	0.021	0.000	0.000					
10	Pr_voice pitch	0.295***	0.110	-0.076	0.098	0.101	-0.100	-0.001	0.174**	0.197**				
11	Pr_fwhr	-0.016	-0.076	0.096	0.051	0.057	0.188**	-0.012	-0.003	0.133	-0.105			
12	Pitch_sent	-0.126	-0.002	-0.119	-0.013	0.002	-0.059	0.032	0.159**	-0.118	0.001	-0.095		
13	Av_fcst_rev_g	-0.050	0.184**	-0.097	0.248***	0.034	-0.012	-0.025	-0.088	0.007	0.075	-0.031	0.079	
14	Breakeven	-0.065	-0.330***	-0.454***	-0.099	-0.008	-0.134*	-0.136*	0.046	-0.048	0.028	0.033	0.100	-0.007

This reports full correlation matrix for the main variables in the study (note: we take the first-year observation for time-variant variables). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

4. Empirical Results and Discussion

To test the effect of entrepreneurs' physical expansiveness on our dependent variables, we estimate the following pooled ordinary least squares (OLS) and logit (where applicable) specification:

$$\text{Outcome variable} = \alpha + \text{Pers_char}\beta + X\eta + \delta + \gamma + \varepsilon. \quad (1)$$

Outcome variable is one of the dependent variables (i.e., revenue and income forecast error for firm i in year t , a high firm valuation indicator variable at forum date, survival, and investment outcome). *Pers_char* is our variable of interest, which includes the physical expansiveness measure (*Expansiveness*) and any of the three personal characteristics (*Presence*, *Attractive*, and *Professional*). X represents the control variables. The model also includes year and industry fixed effects (δ and γ , respectively) and clustered standard errors at industry level (or double clustered at industry and firm level when possible). Appendix B describes all variables.

4.1 PHYSICAL EXPANSIVENESS

Our first research question explores the associations between physical expansiveness and revenue and earnings forecast errors and overly high valuation numbers (as compared to round peers). Panel A of table 5 uses the full sample (i.e., presentations including one or multiple speakers) and reports a positive and significant association between *Expansiveness* and revenue forecast errors (coefficient of 0.158 and t -statistic of 2.63). The magnitude of the coefficient represents an increase of about 4.5% of the average revenue forecast error for every unit (standard deviation) increase in *Expansiveness*. Likewise, more expansive individuals also report higher earnings forecast errors (coefficient of 0.211 and t -statistic of 5.02), which is equivalent to an increase of about 6.4% for every standard deviation increase in physical expansiveness. Moreover, physically expansive entrepreneurs are more likely to set valuations with multiples over revenues exceeding the mean multiple of its round peers (*Val_indic* coefficient of 1.010 and t -statistic of 2.94). Results reported in panel B of table 5 remain robust when restricting the sample to one-speaker presentations (i.e., no aggregation of physical expansiveness across multiple speakers).^{10,11}

¹⁰ Alternative measures of forecast errors and overly high firm valuations are presented in online appendix 4. We also corroborate that results are robust to controlling for speaker's gender.

¹¹ In untabulated tests, we first explore an interaction effect between *Expansiveness* and forecasted information (*Av_fcst_rev_g*) in the *Invest* specification (column 5). We observe that the individual coefficient of *Expansiveness* remains significant at a 10% level (1% level in the one-speaker sample), meaning that, irrespective of the optimism reflected in forecasted information, body expansiveness increases the likelihood of funding success. These investor perceptions of entrepreneurs are marginally strengthened with more optimistic forecasts (i.e., higher annualized revenue growth rates), suggesting that the positive qualities investors place on entrepreneurs become more prevalent with a more demanding business plan.

TABLE 5
Physical Expansiveness and Outcome Variables

Panel A: Full sample (all projects)					
	(1) Forecast_ Rev_Err	(2) Forecast_ Inc_Err	(3) Val_indic	(4) Active	(5) Invest
Expansiveness	0.158*** (2.63)	0.211*** (5.02)	1.010*** (2.94)	−0.360** (−2.46)	0.280*** (3.19)
Firm age	0.001 (0.72)	−0.002 (−1.26)	−0.045*** (−6.66)	0.004* (1.69)	−0.001 (−0.25)
Employees	−0.013*** (−4.34)	0.014** (2.20)	−0.409*** (−4.15)	−0.081** (−2.40)	0.094* (1.81)
Venture Capital	−0.039 (−0.30)	−0.010 (−0.04)	−0.597 (−0.37)	0.181 (0.36)	1.687*** (4.36)
Business Angel	0.032 (0.14)	0.301*** (3.66)	−0.953*** (−4.49)	0.107 (0.24)	0.387 (1.01)
Accelerator	−0.203 (−0.37)	0.174 (0.45)	−0.372 (−0.77)	−0.899 (−1.64)	−1.595*** (−4.69)
Debt	0.010 (0.05)	0.014 (0.11)	−0.023 (−0.04)	0.125 (0.16)	1.321*** (3.71)
Log(Capital)	−0.002 (−0.04)	−0.073 (−0.66)	2.066*** (3.43)	0.549** (2.34)	−0.248 (−0.88)
Round number	−0.236*** (−4.23)	−0.240*** (−2.70)	0.368 (0.74)	0.057 (0.29)	0.129 (0.69)
Male ratio	−0.043 (−0.13)	−0.535*** (−3.05)	0.600 (0.76)	−2.194*** (−2.85)	1.962* (1.66)
Industry experience ratio	0.010 (0.13)	0.376** (2.60)	−1.234 (−1.30)	0.998*** (3.43)	1.197 (1.31)
Startup experience ratio	−0.456** (−2.41)	−0.846*** (−2.71)	−2.305*** (−5.08)	−0.139 (−0.41)	−1.897** (−2.47)
Pr_voice pitch	−0.001 (−0.32)	−0.007*** (−4.70)	0.003 (0.28)	0.008** (2.45)	−0.012 (−1.30)
Pr_fWHR	−0.101*** (−3.43)	−0.078 (−1.43)	0.205 (0.69)	−0.010 (−0.03)	0.222*** (3.09)
Pitch_sent	−1.898 (−1.55)	−1.663 (−1.19)	3.064 (0.40)	−2.116 (−0.37)	0.015 (0.01)
Av_fcst_rev_g					0.000 (0.56)
Breakeven					−0.282*** (−3.26)
Year FE	Y	Y	Y	Y	Y
Industry FE	Y	Y	Y	Y	Y
Model	OLS	OLS	Logit	Logit	Logit
Observations	441	441	136	154	154
R ² (OLS) or Pseudo R ² (Logit)	0.256	0.312	0.382	0.267	0.265

(Continued)

TABLE 5—(Continued)

Panel B: Projects featuring one speaker					
	(1) Forecast_ Rev_Err	(2) Forecast_ Inc_Err	(3) Val_indic	(4) Active	(5) Invest
Expansiveness	0.174** (2.61)	0.216*** (4.81)	0.963*** (4.01)	−0.400** (−2.41)	0.256*** (3.95)
Firm age	0.002 (1.22)	−0.002* (−1.97)	−0.063*** (−5.78)	0.003** (2.13)	−0.000 (−0.02)
Employees	−0.019** (−2.17)	0.026*** (4.61)	−0.379*** (−2.85)	−0.097*** (−4.22)	0.069 (1.05)
Venture Capital	−0.020 (−0.15)	−0.009 (−0.03)	−0.816 (−0.74)	−0.111 (−0.29)	1.359*** (2.67)
Business Angel	0.092 (0.41)	0.453*** (10.06)	−1.416*** (−3.22)	−0.149 (−0.43)	−0.153 (−0.27)
Accelerator	−0.169 (−0.27)	0.357 (0.94)	−0.783 (−0.67)	−1.111** (−2.28)	−2.145*** (−3.09)
Debt	−0.088 (−0.96)	−0.099 (−1.17)	0.599 (1.12)	0.488 (0.52)	0.870** (1.97)
Log(Capital)	0.011 (0.25)	−0.110 (−1.05)	2.184*** (3.84)	0.576** (2.01)	−0.208 (−0.83)
Round number	−0.289*** (−2.72)	−0.360*** (−6.30)	0.525** (2.30)	0.140 (0.53)	0.326 (1.09)
Male ratio	0.080 (0.24)	−0.386** (−2.45)	0.477 (0.49)	−2.319*** (−3.03)	1.758** (1.99)
Industry experience ratio	−0.083 (−1.51)	0.337 (1.19)	−1.263 (−1.56)	0.886*** (3.41)	0.901 (0.97)
Startup experience ratio	−0.477** (−2.34)	−0.776** (−2.58)	−2.611*** (−3.33)	−0.076 (−0.16)	−1.607*** (−2.72)
Pr_voice pitch	−0.001 (−0.15)	−0.007*** (−3.20)	0.012 (0.72)	0.010*** (5.69)	−0.006 (−0.71)
Pr_fWHR	−0.108** (−2.28)	−0.057 (−0.87)	0.209 (0.63)	−0.053 (−0.18)	0.189*** (2.86)
Pitch_sent	−1.443 (−1.15)	−1.550 (−1.43)	0.161 (0.02)	−1.736 (−0.23)	−0.643 (−0.28)
Av_fcst_rev_g					0.000 (0.22)
Breakeven					−0.309*** (−2.66)
Year FE	Y	Y	Y	Y	Y
Industry FE	Y	Y	Y	Y	Y
Model	OLS	OLS	Logit	Logit	Logit
Observations	394	394	122	137	137
R ² (OLS) or Pseudo R ² (Logit)	0.256	0.354	0.424	0.275	0.231

(Continued)

TABLE 5—(Continued)

Panel A of table 5 reports OLS regression coefficients from estimating equation (1) with *Forecast_Rev_Err* and *Forecast_Inc_Err* as dependent variables (columns 1 and 2). These specifications use the overall pooled sample of yearly forecasts (with standard errors clustered at industry and firm level). Columns 3 to 5 report Logit specifications for the *Val_ind*, *Active*, and *Invest* outcome variables (standard errors clustered at industry level). *Val_indic* shows only 136 observations because of some entrepreneurs not reporting their company valuations (missing values). *Forecast_Rev_Err* is the revenue forecast error (classified into quantiles), which compares forecasted to actual revenue numbers and scale it by total assets. *Forecast_Inc_Err* is the income forecast error (classified into quantiles), which compares forecasted to actual income numbers and scale it by total assets. *Val_indic* takes the value of 1 when proposed firm valuations are above the mean valuation (over revenues) for a given round. *Active* is an indicator variable that takes the value of 1 for surviving firms. *Invest* is an indicator variable that takes the value of 1 when the company successfully raised financing. *Expansiveness* is the objective measure of physical display (see section 3.2 on paper for further details). Panel B of table 5 reports OLS regression coefficients from estimating equation (1) with *Forecast_Rev_Err* and *Forecast_Inc_Err* as dependent variables (columns 1 and 2). These specifications use the overall pooled sample of yearly forecasts of 137 projects presented by just one entrepreneur (speaker). We cluster standard errors at industry and firm level. Columns 3 to 5 report Logit specifications for the *Val_ind*, *Active*, and *Invest* outcome variables (standard errors clustered at industry level). *Val_indic* shows only 122 observations because of some entrepreneurs not reporting their company valuations (missing values). *Forecast_Rev_Err* is the revenue forecast error (classified into quantiles), which compares forecasted to actual revenue numbers and scale it by total assets. *Forecast_Inc_Err* is the income forecast error (classified into quantiles), which compares forecasted to actual income numbers and scale it by total assets. *Val_indic* takes the value of 1 when proposed firm valuations are above the mean valuation (over revenues) for a given round. *Active* is an indicator variable that takes the value of 1 for surviving firms. *Invest* is an indicator variable that takes the value of 1 when the company successfully raised financing. *Expansiveness_Ispeak* is the objective measure of physical display (see section 3.2 on paper for further details). Appendix B defines all variables. ^{*} $p < 0.10$, ^{**} $p < 0.05$, ^{***} $p < 0.01$.

As noted above, the social psychology literature associates more expansive behaviors, such as widespread limbs or the maximization of occupied space, with dominant, passionate, and attractive individuals (Carney, Hall, and LeBeau [2005], Chen, Yao, and Kotha [2009], Vacharkulksemsuk et al. [2016]). Vallerand et al. [2003] conclude that passionate individuals tend to be more obsessive and misdirected toward a narrow view of the venture. Kahneman and Lovallo [1993] term this shortsighted approach the “inside view” and claim that individuals who overly rely on the specifics of a case at hand (i.e., the company) tend to limit their search for information, resulting in “rosy” expectations and a higher likelihood of higher forecast errors and overvaluations. Characteristics related to dominant individuals may also explain the reported results: these people tend to possess higher feelings of power and control (Fast et al. [2009]), which manifest in overconfidence (Fast et al. [2012]). Likewise, attractive individuals have been found to develop better social skills, which stems from social success from as early as childhood, when teachers devoted more attention to them (Hatfield and Sprech [1986], Heckman [2000]). This preferential treatment builds confidence (Mobius and Rosenblat [2006]), which later can raise expectations about a firm’s prospects (i.e., higher forecast errors and firm valuations).

Our second research question explores whether entrepreneurs’ physical expansiveness is associated with firm survival and actual investor funding decisions. We find that firms led by more expansive entrepreneurs are less likely to survive (panels A and B of table 5, column 4). More dominant individuals have a higher tolerance for risk (Anderson and Galinsky [2006]) and are overconfident about their firm’s ability to generate returns

(Malmendier and Tate [2008]). Hence they are more prone to decisions that can lead to monetary losses (Fast et al. [2012]). Altogether, it seems plausible that the personal characteristics associated with more expansive displays make individuals more prone to take actions (i.e., burn cash) that harm the long-run survival of the firm.

Paradoxically, investors are more likely to fund companies managed by more expansive entrepreneurs (panels A and B of table 5, column 5). We argue that dominant and passionate individuals are perceived to be associated with positive outcomes, like tenacity in pursuing goals, inspirational leadership, and dedication to the venture (Murnieks et al. [2016]), qualities that attract investors. Overall, it seems investors do not correctly incorporate the implications (e.g., lower survival) of these types of entrepreneurs' characteristics. We provide analysis on alternative (rational) explanations in section 4.3.1.

The effect of various control variables is worth mentioning. First, a governance measure, like number of previous financing rounds (which implicitly identifies the presence of professional investors), is negatively correlated to forecasting errors (panels A and B of table 5, columns 1 and 2). This finding suggests that better governed firms (e.g., with proper control mechanisms) report more accurate information. We further develop this argument when exploring heterogeneous effects of physical expansiveness in section 4.4. Second, debt relates positively to funding success (panels A and B of table 5, column 5), which can be explained by both higher availability of cash and governance mechanisms imposed by lenders (Epure and Guasch [2020]). Finally, entrepreneurs' characteristics also seem important, with projects led by a more male entrepreneurial team showing higher likelihood of funding success (Brooks et al. [2014]), yet being less likely to survive (panels A and B of table 5, columns 4 and 5). Startup experience appears significant for producing lower forecasting errors and less overstatement of valuation numbers; however, entrepreneurs with this experience are less likely to receive investments, which suggests that investors associate past failures with lack of management skills. Finally, voice pitch is negatively associated with income forecast error but positively with survival, whereas $fWHR$ is positively correlated with investing, which is consistent with prior work. Less time to break even is negatively related to receiving an investment, suggesting that investors perceive a quick path to profitability as signaling lower potential multiples.

4.2 INVESTOR PERCEPTIONS OF ENTREPRENEURS

Our paper builds on the premise that more physical expansiveness is associated with more dominant, passionate, and attractive individual characteristics, which ultimately affect firm outcomes. In this section, we provide empirical support for our main argument using an alternative measure based on ratings (from an independent group of raters) of entrepreneurs' dominance, passion, and attractiveness. Results consistent with those for our *Expansiveness* measure reinforce the proposed underlying mechanism.

TABLE 6
Descriptive Statistics Turk Prime Respondents

	Frequency	Percent
Gender		
Male	744	41.99
Female	1,028	58.01
Total	1,772	100.00
Working experience		
< 10 years	470	26.52
1–29	711	40.12
≥30	591	33.35
Total	1,772	100.00
Age		
<30 years old	307	17.33
30–49	682	38.48
≥50	783	44.19
Total	1,772	100.00
Education		
Some high school but no diploma	27	1.52
High school graduate or equivalent	277	15.63
Trade, technical, or vocational training	175	9.88
Some college but no diploma	378	21.33
College graduate	611	34.48
Some postgraduate work but no diploma	61	3.44
Postgraduate degree	243	13.71
Total	1,772	100.00
Ethnicity		
African American	97	5.47
Asian	41	2.31
Caucasian (White)	1,490	84.09
Hispanic	103	5.81
Other	41	2.31
Total	1,772	100.00

Table 6 provides demographic characteristics of 1,772 respondents that successfully answer the survey.

A second benefit of analyzing perceptions is that they more resemble an actual investment decision, one of our outcome variables of interest; after all, the decision to invest is made to a great extent based on perceptions of the entrepreneur.

Table 6 provides demographic characteristics of the (valid) 1,772 respondents. They are mostly female (58%), have between 10 and 29 years of work experience (40%), and are older than 50 years (44%). In addition, 84% of respondents are Caucasian (white), and about half of the sample holds a college degree or more (52%), a statistic that resembles other studies using independent raters (e.g., Blankespoor, Hendricks, and Miller [2017]).

From the survey, we obtain five measures per entrepreneur, which are highly correlated. We reduce the dimensionality of the original variables to three components (see online appendix 3 and section 3.3): *Presence*

TABLE 7
Perceptions of Entrepreneurs and Outcome Variables

	(1) Forecast_ Rev_Err	(2) Forecast_ Inc_Err	(3) Val_indic	(4) Active	(5) Invest
Presence	0.101*** (4.06)	0.089** (2.01)	0.269*** (2.62)	−0.329*** (−2.58)	0.664** (2.28)
Attractive	0.177*** (3.72)	0.139** (2.28)	−0.110 (−0.37)	−0.424 (−1.55)	0.292 (1.13)
Professional	−0.005 (−0.06)	−0.049 (−0.81)	0.274 (0.63)	0.101 (0.61)	0.289 (0.74)
Controls	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y
Industry FE	Y	Y	Y	Y	Y
Model	OLS	OLS	Logit	Logit	Logit
Observations	441	441	136	154	154
R ² (OLS) or Pseudo R ² (Logit)	0.266	0.309	0.333	0.280	0.304

Table 7 reports OLS regression coefficients from estimating equation (1) with *Forecast_Rev_Err* and *Forecast_Inc_Err* as dependent variables (columns 1 and 2). These specifications use the overall pooled sample of yearly forecasts (with standard errors clustered at industry and firm level). Columns 3 to 5 report Logit specifications for the *Val_ind*, *Active*, and *Invest* outcome variables (standard errors clustered at industry level). *Val_indic* shows only 136 observations because of some entrepreneurs not reporting their company valuations (missing values). *Forecast_Rev_Err* is the revenue forecast error (classified into quantiles), which compares forecasted to actual revenue numbers and scale it by total assets. *Forecast_Inc_Err* is the income forecast error (classified into quantiles), which compares forecasted to actual income numbers and scale it by total assets. *Val_indic* takes the value of 1 when proposed firm valuations are above the mean valuation (over revenues) for a given round. *Active* is an indicator variable that takes the value of 1 for surviving firms. *Invest* is an indicator variable that takes the value of 1 when the company successfully raised financing. *Presence*, *Attractive*, and *Professional* have been constructed after a PCA on the independent ratings from Turk Prime participants. *Presence* captures dominance and passionate personal characteristics, *Attractiveness* captures ratings on attractiveness alone, whereas the *Professional* component bundles trustworthiness and competence. Section 3.3 on paper provides further details on the survey and PCA variables reduction process. Appendix B defines all variables. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

(capturing dominance and passion), *Attractive*, and *Profession* (trustworthiness and competence). Table 7 reports the association between the three components and the five outcome variables of interest. *Presence* is positively and significantly associated with revenue and earnings forecast errors (coefficient of 0.101 and 0.089 and t -statistics of 4.06 and 2.01, respectively). The magnitude of the coefficients represents an increase of about 2.7%–2.8% of the average forecast error for every unit (standard deviation) increase in *Presence*. We further find that *Presence* is positively associated with high firm valuations. It is perhaps puzzling to associate perceptions of entrepreneurs with outcomes related to their information preparation processes—firm forecasting errors and overly high proposed valuations. We rely on findings from social psychology, which suggest that perceptions of others—captured through dynamic physical displays—are indeed accurate approximations of *true* individual characteristics (Ambady and Rosenthal [1992], Ambady, Bernieri, and Richeson [2000]). To the extent investors accurately capture the characteristics of dominance, passion, and attractiveness, then

the effects of their perceptions on firm reporting and performance become justified.

Table 7 also reports a negative association between *Presence* and firm survival. Analogous to our physical expansiveness arguments, we attribute these findings to passionate and dominant individuals tolerating more risk (Anderson and Galinsky [2006]) and being more overconfident about their firm's ability to generate returns (Malmendier and Tate [2008]), hence making decisions leading to losses (Fast et al. [2012]). Altogether, it seems plausible that individuals with greater presence are more ready to take actions (i.e., burn cash) that can harm the firm in the long run. Yet, despite all the above, investors are more likely to invest in firms run by these types of entrepreneurs.

Attractive is also positively associated with higher revenue and earnings forecast errors (table 7, columns 1 and 2). This is consistent with attractiveness being associated with more overconfidence (Fast et al. [2012]) and higher self-confidence (Hatfield and Sprecher [1986], Mobius and Rosenblat [2006]), which prompt more optimistic forecasts. The control variable *Professional* is insignificant.

A valid concern is whether these perceptions accurately capture the constructs we interpret (e.g., *Presence* capturing dominance and passionate) or instead confounding omitted variables. For instance, extraversion tends to be manifested in outgoing and energetic behavior (Green, Jame, and Lock [2019]), patterns that could also be found in more dominant and passionate people. One feature of extraverted individuals is that they speak using more positive and negative emotion words (Green, Jame, and Lock [2019, p.178]). We examine this possibility through the use of the R Syuzhet package, which provides, besides traditional sentiment scores, the number of words of eight basic emotions (anger, anticipation, disgust, fear, joy, sadness, surprise, and trust). In an untabulated analysis, we control for these emotion words, and results (from table 7) remain unaltered. More generally, although establishing boundaries of personality or psychological constructs is very challenging, we believe the inclusion of pitch sentiment as a control variable increases the robustness of our results to other potential confounding individual traits.

Overall, results from the perception analysis are very much aligned with those obtained in the main analysis, providing empirical comfort in at least two ways. First, it reinforces the validity of our objective measure of expansiveness, and, second, it provides greater confidence in our proposed explanation.

4.3 RULING OUT ALTERNATIVE ARGUMENTS

4.3.1. Rational Investments in Search for High Returns. One of our main results is based on a behavioral argument that investors do not correctly assess the informational value of nonverbal communication. They fail to understand the implications of investing with more expansive entrepreneurs, who report more biased forecasts, seek higher valuations, and

whose firms are less likely to survive. We explain this anomaly by arguing that individuals (investors) associate positive qualities with more passionate, dominant, and attractive others (entrepreneurs), even when these qualities do not predict better performance (e.g., survival).

There is, however, a plausible alternative (rational) explanation to the observed investment behavior. Investors may prefer expansive entrepreneurs because they seek high returns that are more likely to materialize from managers who communicate qualities like assertiveness, tenacity, or risk tolerance (i.e., qualities associated with dominant, passionate, and attractive individual characteristics). Early-stage investments are structured in such a way that one project (the “home run”) can compensate for a majority of loss-making investments, so investors seek projects that may offer high growth (but often high risk too). To further disentangle the behavioral versus rational explanations, we perform a series of quantitative and qualitative analyses.

First, we empirically test ex post performance. In line with Graham, Harvey, and Puri [2017], we argue that, if a rational argument were to prevail, we should observe that more expansive entrepreneurs—those investors prefer to rely on—deliver better performance, which ultimately translates into higher exit values and returns (i.e., MOICs and IRRs). Unfortunately, we cannot observe exit values or actual MOICs and IRRs.¹² Instead, we can analyze revenue growth as a proxy for exit opportunities. In particular, we test three different growth measures: *Rev_g* (firm-level yearly revenue growth in absolute € amount), *Rev_g%* (firm yearly revenue growth in percentage), and *Rev_gAcc* (the accumulated € amount of revenue growth over the sample years). Panel A of table 8 reports summary statistics for these measures.

Panel B of table 8 reports regression results of growth on *Expansiveness*.¹³ Columns 1 to 3 report results on the full sample whereas columns 4 to 6 display the regression output conditional on survival. None of the six specifications shows positively significant results—in fact, columns 2 and 5 display a negative association between *Expansiveness* and *Rev_g%*. We also evaluate these specifications at the 75th quantile of the growth distribution (i.e., capturing the effect of *Expansiveness* on growth using observations closer to the home runs). Untabulated results present insignificant results.

We also reproduce the above analysis with employee growth, and conclusions remain unaltered (i.e., all coefficients remain insignificant). Additionally, we test for profitability measures that can be of interest to

¹² Exit values and actual MOICs and IRRs are confidential information that only materializes several years after an investment. It is highly likely that some of our observed investments are not yet even realized (exited). We asked the investment network for additional information on this issue, but unfortunately they do not keep track of presented projects’ future performance or financing outcomes.

¹³ To maximize the number of observations in these performance tests, we retrieve the most recent data from Bureau van Dijk (June 2021), irrespective of satisfactory matchings with the original sample (i.e., forecasted data). Hence the increase in the number of observations to $N = 527$.

TABLE 8
Physical Expansiveness and Growth

Panel A: Descriptive statistics (growth variables)								
Variable name	N	Mean	Std. dev	Min	Max	p25	p50	p75
Revg	527	0.489	4.440	−6.202	88.999	−0.009	0.041	0.191
Rev_g%	527	1.569	7.532	−0.976	102.230	−0.117	0.292	1.060
Rev_gAcc	527	1.084	7.233	−3.189	147.575	0.008	0.105	0.414
Panel B: Regressions of growth on physical expansiveness								
	(1) Rev_g	(2) Rev_g%	(3) Rev_gAcc	(4) Rev_g Active	(5) Rev_g% Active	(6) Rev_gAcc Active		
Expansiveness	−0.116 (−0.77)	−0.288*** (−2.98)	−0.052 (−0.27)	−0.110 (−0.66)	−0.314** (−2.03)	−0.042 (−0.17)		
Controls	Y	Y	Y	Y	Y	Y		
Year FE	Y	Y	Y	Y	Y	Y		
Industry FE	Y	Y	Y	Y	Y	Y		
Model	OLS	OLS	OLS	OLS	OLS	OLS		
Observations	527	527	527	468	468	468		
R ²	0.857	0.045	0.857	0.861	0.059	0.862		

Panel A of table 8 reports statistics for growth measures. *Rev_g* is the firm-level yearly revenue growth in absolute € amount (in millions). *Rev_g%* represents firm yearly revenue growth in percentage (we apply 1% winsorization to accommodate the presence of outliers). *Rev_gAcc* is the accumulated € amount of revenue growth over the sample years (in millions). In these growth tests, we expand our sample to obtain the highest number of (revenue growth) observations from Bureau van Dijk (irrespective of whether we can match them to firm forecasts). As a result, the number of observations increases to $N = 527$. Panel B reports OLS regression coefficients from estimating equation (1) with *Rev_g*, *Rev_g%*, and *Rev_gAcc* as dependent variables (see description in panel A). Columns 1 to 3 include all firms, whereas columns 4 to 6 condition on survival. In these growth tests, we expand our sample to obtain the highest number of (revenue growth) observations from Bureau van Dijk (irrespective of whether we can match them to our original sample, i.e., firm forecasts). As a result, the number of observations increases to $N = 527$. We cluster standard errors at industry and firm level. *Expansiveness* is the objective measure of physical display (see section 3.2 on paper for further details). Appendix B defines all variables. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

early-stage investors: return on sales (meaningful if the venture is later valued with EBIT/EBITDA multiples), return on equity (a direct measure of investors’ returns), and breakeven after the forum date (positive income numbers likely translate into higher valuations at exit). Consistent with the growth tests, *Expansiveness* coefficients remain insignificant in all of these specifications. Overall, results from these analyses do not seem to support the rational argument that investors bet on expansive entrepreneurs in search of higher expected performance and therefore returns.

We also take a qualitative approach and interview 10 investors in the network. If our expansiveness results reflected the rational/risk argument, we would expect investors to weight entrepreneurs’ characteristics (e.g., attractiveness or passion) that more likely translate into winning investments and therefore returns at exit. We would also expect them to understand that those higher expected returns are associated with higher (expected) risks.

We selected a diverse set of investor types to capture different approaches to startup investing. We interviewed five business angels (all of them with at least seven years of startup investing experience), two family offices, two venture funds, and one corporate fund. These investors differ in their portfolio compositions and objectives. Angels only invest directly in startups. Family offices had startups as one type of asset, usually making them a small part of a portfolio that also included private equity, hedge, and VC funds. Venture funds had their portfolios biased toward later rounds. Corporate investors saw startup investing as a way to explore new value propositions. None of these investors mentioned a portfolio approach to investing, in the sense of consciously using a risk-return criteria to decide whether to invest in a specific startup or in building their portfolio of startups. Rather, they mostly focused on the content of the presentation—market, competitive advantage, differentiation, forecasts, and valuation. When prompted, the interviewees did refer to the presenter's style and mentioned the commitment to the project shown during the presentation. More sophisticated investors (with formalized decision-making processes) did not mention any criteria associated with the style or communication skills of the entrepreneurs. Criteria typically included the business proposition (market, value proposition, differentiation, competition, or competitive advantage) and background of the team. Just one of the interviewees indicated that he was careful with presenters who came across as too aggressive and sales oriented.

In sum, interviews suggest early-stage investors have no explicit risk-return considerations (perhaps because startup investing is already so risky in itself). Investors have a set of criteria—most of them related to market, product, team, and, in some cases, industry—that they benchmark projects against. They focus on the content of presentations, documents, and meetings; they pay little (conscious) attention to the delivery of the content, although, when prompted, they mentioned they considered the style of the presentation. Overall, the interviews did not uncover any argument consistent with a conscious evaluation of body expansiveness or an explicit evaluation of the risk-return profile of their portfolio of investments.

4.3.2. Reverse Causality. Our main results rely on the premise that non-verbal behavior often originates unconsciously (Hall, Coats, and LeBeau [2005]). We acknowledge, however, that our pitch setting does not capture the spontaneity of ordinary social interactions and that a more *discretionary* or artificially planned (nonverbal) behavior—not capturing the true entrepreneur's personal characteristics—could dominate the expansiveness measure we observe.¹⁴ The reverse causality concern would emerge if an

¹⁴ As mentioned in section 3.1, we only visualize entrepreneurs in uninterrupted pitch sessions; data on more spontaneous settings (e.g., Q&A with investors) are not systematically available.

entrepreneur were to rehearse expansive gesturing—for example, to elicit an image of control or confidence—in more inflated or difficult to attain forecasts and valuation numbers.

We take both a qualitative and quantitative approach to address this reverse causality concern. First, we interviewed 10 presenters to better understand their pitch preparation and the reported financial information. Interviewees indicated that they practiced their pitches with their team or videotaped themselves. Yet, most of the feedback was on content, flow, and arguments; they did not recall comments related to their voice or gestures. Furthermore, all presenters were intimately involved in preparing financial information. Overall, the consistency of the 10 interviewees suggests that reverse causality, where optimistic financials drive body expansiveness, does not seem to play a major conscious role. Nonetheless, 10 interviews is a small sample, and we acknowledge these respondents may be affected by, for example, recall bias and sense-making. Thus, we provide a complementary set of quantitative analysis.

We also exploit a specific feature in entrepreneurial pitches: forecasted and valuation information is presented in the last portion of the pitch. We argue that, if entrepreneurs are ready to behave more expansively, because of more inflated forecasts and valuation numbers (i.e., reverse causality), this behavior should be observed in the presentation of the financial information. By removing this portion of the pitch, we attempt to capture a more truthful measure of *innate* body expansiveness. We also remove the initial part of the pitch, which is likely more rehearsed. The goal is to show that our results are robust when we only use the more *innate* nonverbal component. We recompute our Expansiveness measure with the 40%–80% portion (i.e., frames) of the pitch and reproduce our table 5.

Table 9 shows how, even when cutting the *Expansiveness* measure to the arguably more *innate* component, main results remain unaltered.¹⁵ Together, both analyses support our argument that body expansiveness can be traced to entrepreneurs' personal characteristics, which in turn can affect firm outcomes.

4.4 HETEROGENEOUS EFFECTS: THE ROLE OF CORPORATE GOVERNANCE

Our results are likely more prevalent in particular circumstances. We argue that, when corporate governance mechanisms are in place, the effect of physical expansiveness on forecast biases dissipates. Better governance can

¹⁵ We tried different combinations of pitch portions (i.e., frame ranges), including starting from the beginning (0%–80% of frames) or changing the range of the portion we keep (20%–80% or 25%–75%) to compute the more *innate* body expansiveness measure. Results remain unaltered. We also compare expansiveness in the front 80% of the presentation and the back 20% (when financials are presented) and split the sample by forecast optimism. We observe no differences in body expansiveness levels across the pitch (and sample splits), which suggests that entrepreneurs do not use body language to capture investors' attention at specific (more beneficial) parts of the presentation.

TABLE 9
Physical Expansiveness (Partial Pitch Cuts) and Outcome Variables

	(1) Forecast_ Rev_Err	(2) Forecast_ Inc_Err	(3) Val_indic	(4) Active	(5) Invest
Expansiveness (40-80%)	0.138** (2.39)	0.227*** (3.28)	0.869*** (4.47)	-0.370** (-2.50)	0.309*** (4.06)
All other controls	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y
Industry FE	Y	Y	Y	Y	Y
Model	OLS	OLS	Logit	Logit	Logit
Observations	441	441	136	154	154
R ² (OLS) or Pseudo R ² (Logit)	0.253	0.314	0.373	0.268	0.266

Table 9 reproduces the specifications in panel A of table 5 but measuring body expansiveness (*Expansiveness*) using the 40–80% portion of the presentation rather than the entire presentation. The table reports OLS coefficients from estimating equation (1) with *Forecast_Rev_Err* and *Forecast_Inc_Err* as dependent variables (columns 1 and 2). These specifications use the overall pooled sample of yearly forecasts (with standard errors clustered at industry and firm level). Columns 3 to 5 report Logit specifications for the *Val_ind*, *Active*, and *Invest* outcome variables (standard errors clustered at industry level). *Val_indic* shows only 136 observations because of some entrepreneurs not reporting their company valuations (missing values). *Forecast_Rev_Err* is the revenue forecast error (classified into quantiles), which compares forecasted to actual revenue numbers and scale it by total assets. *Forecast_Inc_Err* is the income forecast error (classified into quantiles), which compares forecasted to actual income numbers and scale it by total assets. *Val_indic* takes the value of 1 when proposed firm valuations are above the mean valuation (over revenues) for a given round. *Active* is an indicator variable that takes the value of 1 for surviving firms. *Invest* is an indicator variable that takes the value of 1 when the company successfully raised financing. Appendix B defines all variables. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

prevent highly expansive entrepreneurs’ from impacting reported financial information.

Table 10 shows the coefficient in Expansiveness is positive and significant only in younger companies (less than 27 months old) and those with no presence of VCs.¹⁶ Both variables likely identify better governance, which helps the firm cope with organizational complexities and agency frictions. For older and VC-backed firms, we conclude that the effect of individual personal characteristics—captured through expansive body postures—on forecasting errors is mitigated by stricter governance mechanisms.¹⁷ Moreover, column 6 shows that VC presence motivates less biased (income) forecasts in more expansive entrepreneurs, which might suggest learning dynamics between entrepreneurs and investors.¹⁸ Overall, we document

¹⁶Differences in coefficients across governance (i.e., older and VC-presence) versus no-governance specifications are statistically significant with one exception: *Forecast_Inc_Err* for VC versus non-VC, where the difference of 1.315 (0.241 – (–1.074)) shows a p -value of 0.14. See last row in table 10.

¹⁷We also corroborate that effective governance mechanisms generally attenuate the effect of entrepreneurs’ (perceived) personal characteristics—*Presence*, *Attractiveness*, and *Professional*—on financial forecast errors. See online appendix 5.

¹⁸We cannot reproduce the analysis with the other three outcome variables (valuation, survival, and funding success). Although table 10 builds on yearly forecasts (i.e., continuous mea-

TABLE 10
Heterogeneous Effects: Firm Age and VC Presence

	Firm age				VC presence			
	> Median Firm age		< Median Firm age		Yes		No	
	(1) Forecast_ Rev_Err	(2) Forecast_ Inc_Err	(3) Forecast_ Rev_Err	(4) Forecast_ Inc_Err	(5) Forecast_ Rev_Err	(6) Forecast_ Inc_Err	(7) Forecast_ Rev_Err	(8) Forecast_ Inc_Err
Expansiveness	0.090 (1.24)	-0.040 (-0.31)	0.247*** (4.84)	0.316*** (7.88)	-2.013 (-1.50)	-1.074*** (-3.60)	0.232*** (3.03)	0.241*** (7.13)
Controls	Y	Y	Y	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y	Y	Y	Y
Industry FE	Y	Y	Y	Y	Y	Y	Y	Y
Observations	236	236	205	205	66	66	375	375
R ²	0.354	0.361	0.274	0.407	0.600	0.711	0.272	0.310
Diff in coeff			$\beta_{(3)} - \beta_{(1)}$ **		$\beta_{(4)} - \beta_{(2)}$ ***		$\beta_{(7)} - \beta_{(5)}$ **	
							$\beta_{(8)} - \beta_{(6)}$	

Table 10 reports OLS regression coefficients from estimating equation (1) with *Forecast_Rev_Err* (columns 1, 3, 5, and 7) and *Forecast_Inc_Err* (columns 2, 4, 6, and 8) as dependent variables. Columns 1 to 4 split the sample according to above/below median firm age (27 months), whereas columns 5 to 8 do so according to VC presence (Yes/No) prior to the investment forum. Columns 1 to 4 include all control variables but Firm age (the splitting variable), whereas columns 5 to 8 omit the VC control variable (the splitting variable). *Forecast_Rev_Err* is the revenue forecast error (classified into quantiles), which compares forecasted to actual revenue numbers and scale it by total assets. *Forecast_Inc_Err* is the income forecast error (classified into quantiles), which compares forecasted to actual income numbers and scale it by total assets. *Expansiveness* is the objective measure of physical display (see section 3.2 on paper for further details). Appendix B defines all variables. Standard errors are clustered at firm and industry level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

that effective governance influences the relation between managers' personal characteristics and firm-reported information.

5. Conclusions

Early-stage investors face high uncertainties and information asymmetries that can result in severe agency conflicts with entrepreneurs. To protect against such conflicts, investors demand information that can be used for investment decisions, contract design, and firm monitoring. This paper focuses on the information-collection stage. We analyze whether managers' expansive physical displays relate to forecasting errors and the likelihood of setting overly high firm valuations. We further explore the effect of nonverbal displays on firm survival and actual investment decisions.

We find that physical expansiveness is positively associated with forecasting errors, overly high valuation numbers, and lower survival rates—yet it is also associated with a higher likelihood of receiving equity funding. We argue that the individual characteristics associated with more expansive displays—dominance, attractiveness, and passion—explain our results. Individuals with these characteristics tend to be more confident and assertive (translating into higher forecast errors and firm valuations) and show higher tolerance for risk (translating into lower survival rates).

Investors, however, do not seem to correctly assess the informational value of nonverbal communication. We suggest that the individuals (investors) place a set of positive qualities in others (entrepreneurs), even when these qualities do not predict better performance (e.g., survival, growth, or profitability). Our results are robust to the inclusion of other forms of nonverbal communication (i.e., tone of voice and facial features), which suggests that body language can significantly influence firm outcomes on its own. Overall, our study documents that physical expansiveness, an important yet overlooked component of nonverbal communication, can provide information about financial reporting and firm outcomes.

sure and multiple observations per firm), we only obtain one (binary) observation per firm in the *Val_indic*, *Active*, and *Invest* specifications. We lack statistical power when partitioning the sample by the firm's age and VC presence.

APPENDIX A

Anonymized company profile
COMPANY PROFILE

Company Name	Legal Company name X, Inc.	COMPANY LOGO
Adress, City	Park Avenue, 13, Madrid	
Phone	+34 91 2379 246	
Fax		
Date of establishment		05/06/2012
Project Presenter (name & function)	Name of presenter, Founder and CEO	PRESENTER PICTURE
Contact	Name of contact person	
Phone	+34 624 244 757	
E-mail	email@companyname.com	
DESCRIPTION OF THE BUSSINES IN ONE LINE		
Rent designer dresses and accessories online.		
BUSINESS ACTIVITY - MARKET		
The company rents designer dresses and accessories to Spanish women. These dresses and accessories are mostly worn for formal events such as weddings (160,000 every year), religious events (communions, christenings; 550,000 every year), debuts and other formal events such as cocktails, parties, formal dinners, New Year's Eve, Christmas, etc. We estimate the size of the Spanish market in at least 100m€. The company aims to position itself as Spain's prêt-a-porter rental leader. To achieve this, the company needs to communicate effectively the advantages of renting vs. buying to its 2.5m target public (Spanish women 16–44 who purchase over the internet), as even though this business model has been running in Spain since 2009, customers still are vastly unaware of its existence (in a market survey The company conducted, 57% of respondents responded that they didn't know they could rent their dresses and accessories instead of buying them). We see Spain as the steppingstone for the organic growth of our fashion rental business. Our connections with the fashion and PR industry will enable us to quickly fine-tune our business model, which can be easily exportable to other countries (by teaming up with local fashion players) where middle classes are growing exponentially, there is a growing hunger for fashion and virtually no competition (Mexico, Brazil, Colombia, Chile, Argentina). We are also looking to expand within the EU, to countries such as Italy and the Scandinavian countries.		

VALUE PROPOSAL OF TECHNOLOGY PRODUCT/SERVICE

The company aims to become the Spanish women's closest ally in her "closet full but nothing to wear" moments. Women nowadays are less willing to spend important sums of money on dresses or accessories which they know they will only use in a number of occasions, as they want to wear the latest designs and do not want to repeat the same look twice. The solution of renting these dresses and accessories *ad hoc* for every occasion instead of buying them has become mainstream in the US and UK thanks to businesses like rentthrunway.com (>3.5m users, >\$60m in revenues) and girlmeetsdress.com. We intend to change this at The company; we believe that every woman should be entitled to have her "Cinderella moment", the possibility to wear that dream look for a fraction of its price, and never repeat the same look twice. Our dresses and accessories (which, were you to buy them in the high street would cost up to 4,000€) can be rented from 60€ up to 300€.

At The company you will find the biggest selection of your favorite designers. Renting is as simple as browsing our selection for your choice, selecting your size (we will send you a second size for free, we guarantee that the dress will fit), and the date you want to receive it. Simply receive it comfortably at your home or place of work and return it to our couriers at the end of the rental. We'll take care of the rest!

DISTRIBUTION & MARKETING

Our web is our sole distribution channel. Attractive design, an intuitive interface and safety are our core objectives.

Clients are able to track and receive their orders in the comfort of their homes or their places of work via our partner couriers, who also retrieve the dress once the rental is finished. Our inventory management system monitors SKU rotation and throughput times to ensure inventory optimization.

Our marketing strategy consists of a three-pronged approach, with online advertising (member-get-member, newsletters, SEM/SEO, social media and viral campaigns), print advertising (glossy magazines, airports targeting the "surprise executive dinner or cocktail"...) and physical presentations (pop-up stores in cities where clusters of users form, where our users and their friends can try on our new season collections).

The cost of user registration so far has been €1,4, whereas the margin on the average rental is close to €120. If we assume customers will rent at least 1,5 times over the course of their customer cycle, we are looking at an LTV of €180, and customer acquisition costs of €25/30.

KEY CUSTOMERS

Our biggest client base is retail. However, we are also targeting B2B clients such as fashion stylists, advertising agencies, theater/cinema producers, model agencies and hostess agencies.

KEY ALLIANCES / AGREEMENTS

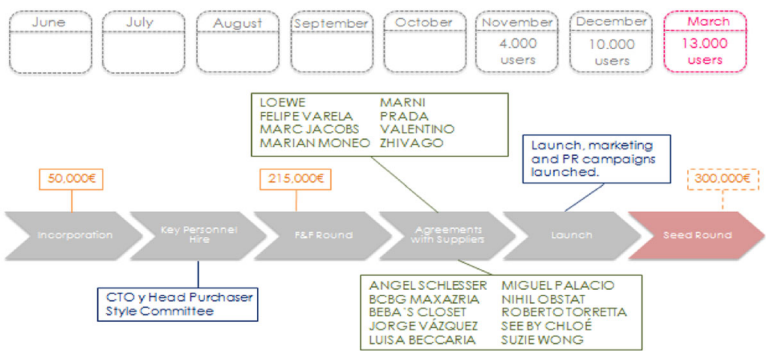
Suppliers: Chloé, Hoss Intropia, Jorge Vázquez, Loewe, Miguel Palacio, Angel Schlessler, Roberto Toretta, Marni, Prada, Marian Moneo, Suzie Wong, Nihil Obstat, Beba's Closet, Valentino, Zhivago, Marc Jacobs, Luisa Beccaria, Gina Foster y Felipe Varela.

IT: Flash2Flash // Legal: Rosaud Costas Duran Abogados y Contalia // Courier: Envialia, ASM.

LEVEL OF COMPETITION & KEY COMPETITORS

A few incumbents have been present in the Spanish market since 2009. The closest business model to The company is 24fab.com. Founded in 2009 by the former CMO of Loewe, 24fab offers dresses and accessory rentals from its showroom in Madrid. Its inventory is listed on their web but customers must either email or call the showroom to inquire availability and process payment. The dress selection is limited in price range (600-1,000€, all rented for a fixed price of 190€) and designers. Their marketing spending is very limited in scope and size. Other smaller competitors are altacosturaonline.com (high-end dress rentals) and mimoki.es (hats and fascinator rentals).

MAIN ACHIVEMENTS/MILESTONES REACHED TO DATE



STRATEGIC DEVELOPMENT GOALS

- 1. To become women’s closest ally in her “closet full but nothing to wear” moments.
- 2. To streamline our operations procedures, reducing throughput times and long-tails.
- 3. To vertically integrate our suppliers in our business model. We are already onboarding new suppliers on a revenue-share scheme.
- 4. To introduce new products and lines of business (LMM Vintage, Insurance schemes, jewelry).
- 5. To expand internationally, Italy and Latam being our priorities.

MANAGEMENT TEAM

Member 1 – CEO: IESE MBA class of 2009. 3 years working in London as country manager for FX products at top-tier investment banks (DB, Nomura). Former pro racing car driver (1996-2003), raised over \$3m in sponsorship. Managed teams of over 20 people and won over 10 international races during his career, as well as 2 Spanish Championships. 7th in the 1996 Karting World Championships.

Member 2 – CTO: Computer engineer, Universidad Complutense Madrid. Owner and founder of Inmasan Proyectos Informáticos S.L., technical consultant for SMEs and start-ups in sectors like finance, leisure, health, real estate, and legal. European runner-up and four-times Spain champion in curling and committee member of the Spanish and Madrid federation.

MANAGMENT TEAM

Member 3 – Head of Purchasing: Fashion Buyer at Unico & Cool, identified and sourced fashion products for the store, organised the logistics and maintenance of stock levels. Product Manager at Pepe Jeans London. Buyer at BCBG Max Azria Group Spain and Don Algodón. Managed the stock selection, international sourcing, logistics and payments.

Member 4 – Head Style and Communication: IEDE MBA and Masters in Fashion Communication at CEU. Long experience in fashion communication and style, having worked for the Press and PR departments of designers like Jesús del Pozo, Kedin.es and Mediterránea Agua de Mar.

ROUNDS OF FINANCING AND INVESTORS

	Date	Investor (name)	Amount (€)	% Share
1st Round	September 2012	Shareholder 1	50,000	65.14
		Shareholder 2	46,600	9.81
		Shareholder 3	30,000	6.31
		Shareholder 4	24,000	5.05
		Shareholder 5	15,000	3.16
		Other shareholders	50,000	10.53

MAIN SHAREHOLDERS

See above

KEY FINANCIAL FIGURES

	Actual data		Forecasts			
	2011	2012	2013	2014	2015	2016
Revenues		€2,200	€160,000	€585,000	€1,291,000	€4,413,000
EBIT		(€65,700)	(€105,500)	(€67,000)	€279,000	€1,740,000
Equity		€135,000	€370,000	€270,000	€1,080,000	€2,120,000
Founders' contribution		€50,000				
External Investors		166.600€	€300.000		€600.000	
Number of employees		3	5	9	12	15

REQUEST EQUITY FINANCING

Total Required funds (euro)	€300,000
Requested equity financing from investors	€300,000
% share for investors	30%
Use of funds	IT: (40,000€) // Inventory: (120,000€) // Marketing: (140,000€) // NOF: 300,000€

OTHER KEY INFORMATION

Links to company's social platforms

APPENDIX B

Variable Definitions

Variable Name	Description
Dependent variables	
Forecast_Rev_Err	Value of [(Forecasts – Actual) / Assets] assigned into 5 quantiles: Lower (higher) quantile represents the more pessimistic (optimistic) revenue forecasts
Forecast_Inc_Err	Value of [(Forecasts – Actual) / Assets] assigned into 5 quantiles: Lower (higher) quantile represents the more pessimistic (optimistic) earnings forecasts
Val_indic	1 = firm valuation is above the mean firm valuation over revenues by financing round; 0 = otherwise
Active	1 = firm is still active (BvD database and manual online inspections); 0 = not active
Invest	1 = firm receives funding from investors in the investment Network; 0 = otherwise
Main independent variables	
Presence	PCA component loading Turk Prime's ratings on passion and dominance
Attractive	PCA component loading Turk Prime's ratings on attractiveness
Professional	PCA component loading Turk Prime's ratings on trustworthiness and competence
Control variables	
Firm age	Age (in months) at forum date
Employees	Number of employees
Venture Capital	Indicator variable that equals 1 with Venture Capital presence; 0 otherwise
Business Angel	Indicator variable that equals 1 with Business Angel presence; 0 otherwise
Accelerator	Indicator variable that equals 1 with business Accelerator presence; 0 otherwise
Debt	Indicator variable that equals 1 with debt presence; 0 otherwise
Log(Capital)	Logarithm of capital solicited to investors
Round number	Financing round number
Male ratio	Percentage of males in entrepreneurial team
Industry experience ratio	Percentage of team members with prior industry experience
Startup experience ratio	Percentage of team members with prior startup experience
Pr_voice pitch	Mean presenter voice pitch fundamental frequency (extracted using Praat software)
Pr_fWHR	Presenter facial weight to height ratio
Pitch_sent	Pitch negative sentiment. Number of negative words over total number of negative and positive words (extracted using R Syuzhet package)
Av_fcst_rev_g	Average forecasted revenue growth (at pith date) over all disclosed forecasted years
Breakeven	Number of years that forecasted income takes to breakeven (i.e., show a positive amount)

Variable Name	Description
Expansiveness—Component resulting from a PCA on:	
Feet motion	Average distance (in number of pixels) between two consecutive right or left foot coordinates
Hands motion	Average distance (in number of pixels) between two consecutive right or left hand coordinates
Head motion	Average distance (in number of pixels) between two consecutive head coordinates

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